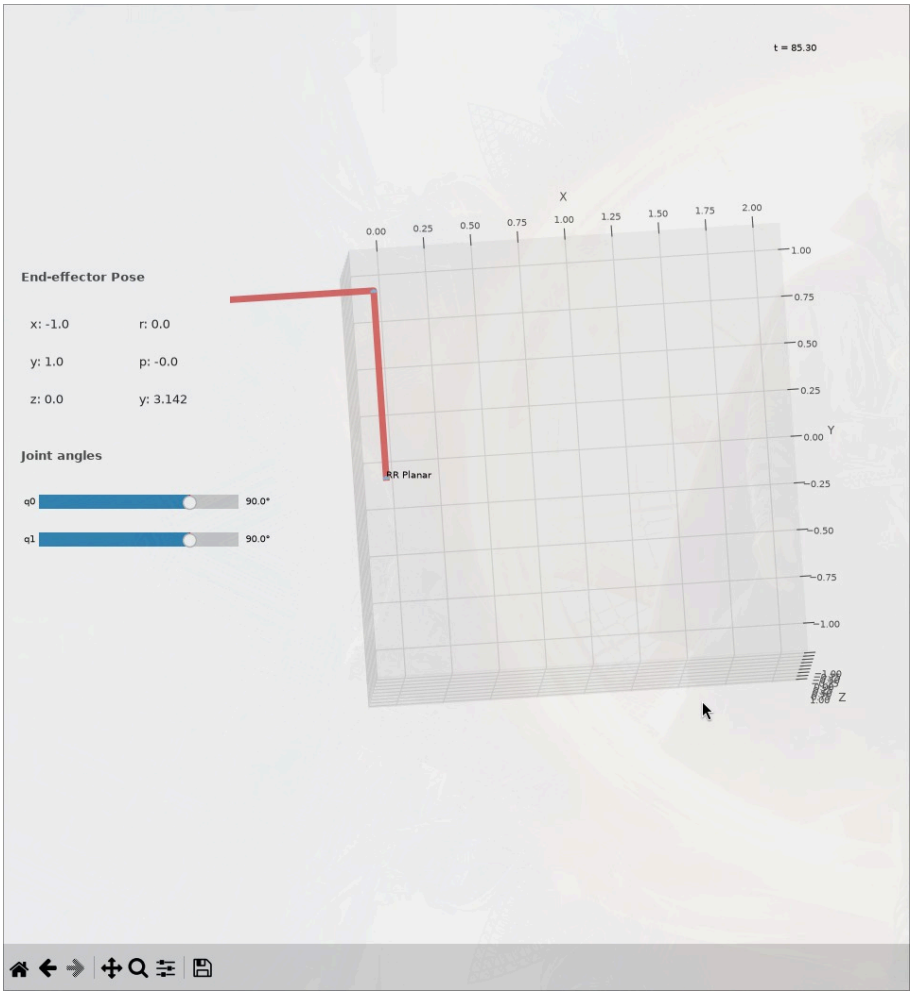
EE430 Laboratory Report 1: Forward and Inverse Kinematics

Part 1: Forward Kinematics - Planar RR Robot Manipulator

- **Implementation:** A custom getTransformationMatrix function was written to compute homogeneous transformation matrices using Denavit-Hartenberg (DH) parameters.
- **Application:** This function was used to calculate the transformation matrix for a 2-DOF planar RR manipulator assuming 1m link lengths. The robot's configuration was verified and visualized using Peter Corke's Robotics Toolbox.



Output:

```
The angles theta_1 = 90.00 theta_2 = 90.00

Transformation Matrix T_0^2:
  -1         0         0        -1
   0        -1         0         1
   0         0         1         0
   0         0         0         1
```

```
-1         0         0        -1
 0        -1         0         1
 0         0         1         0
 0         0         0         1
```

Part 2: Inverse Kinematics - Planar RR Robot Manipulator

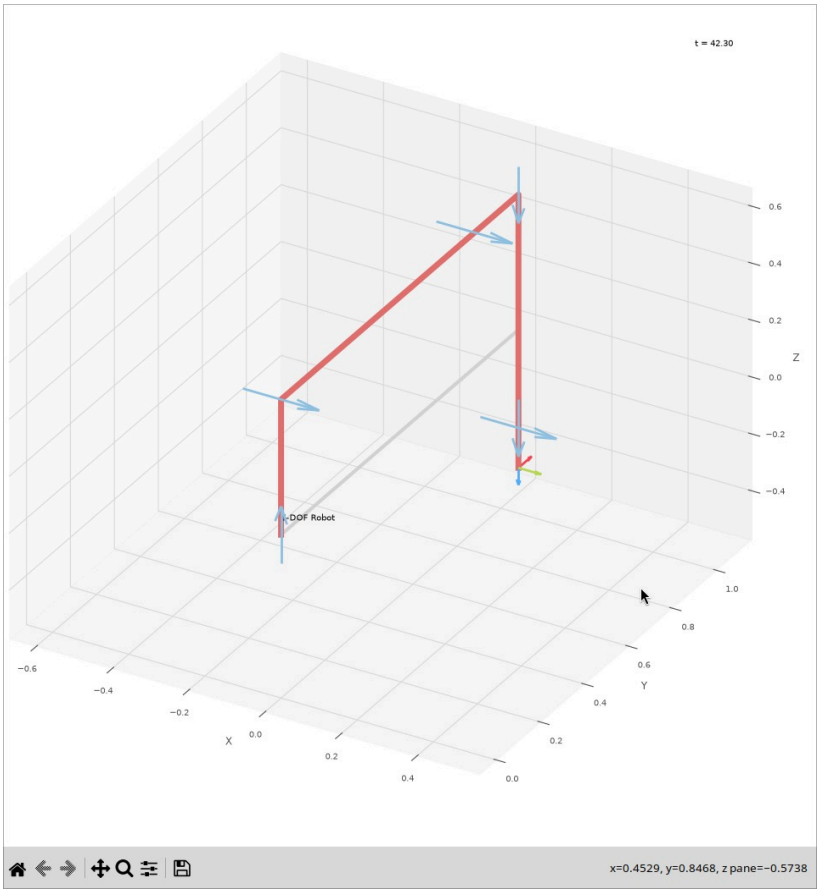
- **Analytical Solution:** An algorithm was developed to calculate the required joint angles **theta_1** and **theta_2** to reach specific 2D coordinates.
- **Error Handling:** The system successfully identifies unreachable coordinates mathematically and prints an error message instead of failing.
- **Verification:** The calculated angles were fed back into the Robotics Toolbox to physically verify that the end-effector reached the target point.

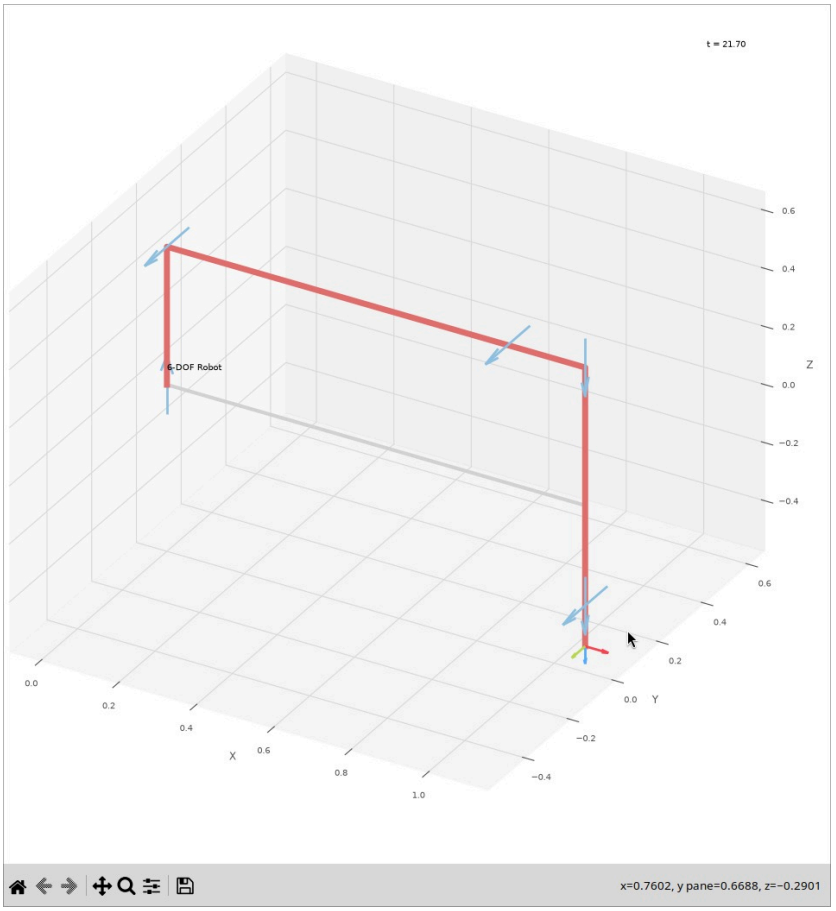
Output:

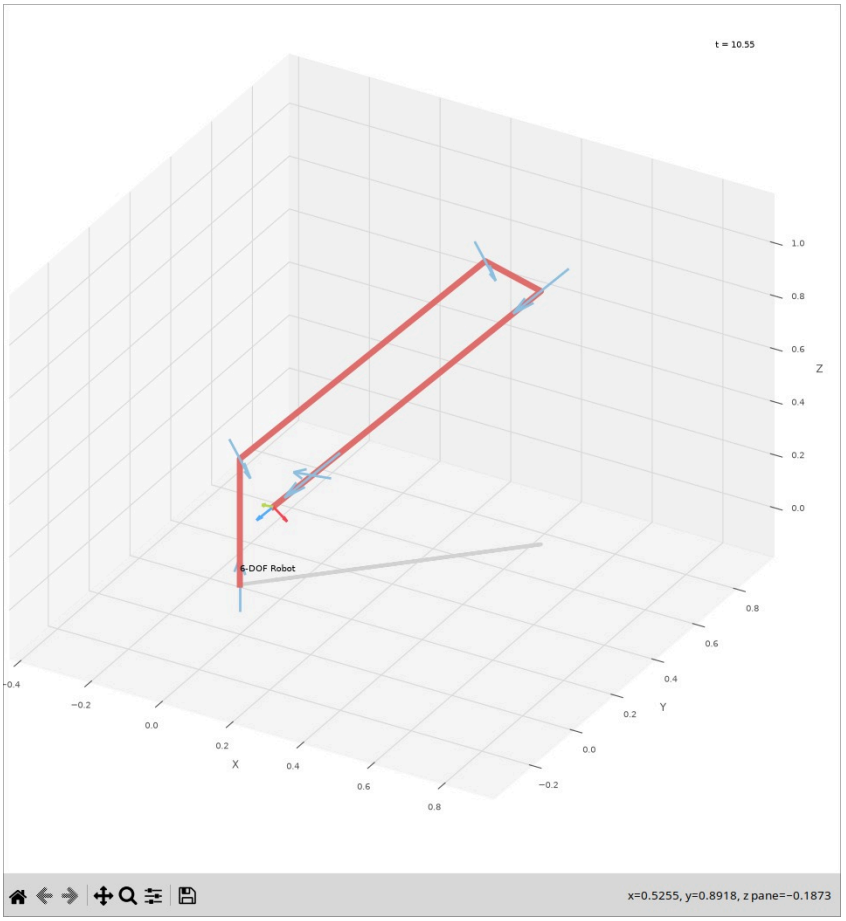
```
Success: Point is reachable.
Calculated Angles -> Theta 1: -19.33°, Theta 2: 75.52°
Target Point:   X = 1.5000, Y = 0.5000
Achieved Point: X = 1.5000, Y = 0.5000
Verification Successful: The manipulator reaches the target point!
Error: The point (3.0, 3.0) is not reachable by the manipulator.
```

Part 3: Group Specific Task (6-DOF Robot)

- **Forward Kinematics:** The custom transformation function was applied to an assigned 6-DOF robot configuration. Matrices for successive frames T_i-1_i and the total transformation matrix T_0_n were calculated and printed for three different poses.
- **Visualization:** The three distinct poses were visualized using the toolbox to confirm the mathematical outputs.
- **Inverse Kinematics:** The inverse kinematics task was repeated for the 6-DOF arm, utilizing the ikine_LM numerical solver to find joint configurations for desired 3D positions and orientations.


t = 42.30
6-DOF Robot
X
Y
Z
x=0.4529, y=0.8468, z pane=−0.5738


t = 21.70
6-DOF Robot
X
Y
Z
x=0.7602, y pane=0.6688, z=−0.2901



Output:

```
T10
   0         0         1         0
   1         0         0         0
   0         1         0         0.475
   0         0         0         1

T21
   1         0         0         0.88
   0         1         0         0
   0         0         1         0
   0         0         0         1

T32
   1         0         0         0.2
   0         0        -1         0
   0         1         0         0
   0         0         0         1

T43
   1         0         0         0
```

```
   0         0         1         0
   0        -1         0         0.82
   0         0         0         1

T54
   1         0         0         0
   0         0        -1         0
   0         1         0         0
   0         0         0         1

T65
   1         0         0         0
   0         1         0         0
   0         0         1         0.14
   0         0         0         1

T60
   0         1         0         0
   1         0         0         1.08
   0         0        -1        -0.485
   0         0         0         1

-----------------------------------------------------------------------
T10
   1         0         0         0
   0         0        -1         0
   0         1         0         0.475
   0         0         0         1

T21
   1         0         0         0.88
   0         1         0         0
   0         0         1         0
   0         0         0         1

T32
   1         0         0         0.2
   0         0        -1         0
   0         1         0         0
   0         0         0         1

T43
   1         0         0         0
   0         0         1         0
   0        -1         0         0.82
   0         0         0         1
```

```
T54
   1         0         0         0
   0         0        -1         0
   0         1         0         0
   0         0         0         1

T65
   1         0         0         0
   0         1         0         0
   0         0         1         0.14
   0         0         0         1

T60
   1         0         0         1.08
   0        -1         0         0
   0         0        -1        -0.485
   0         0         0         1

-----------------------------------------------------------------
T10
   0.7071    0         0.7071    0
   0.7071    0        -0.7071    0
   0         1         0         0.475
   0         0         0         1

T21
   0.7071   -0.7071    0         0.6223
   0.7071    0.7071    0         0.6223
   0         0         1         0
   0         0         0         1

T32
   0         0        -1         0
  -1         0         0        -0.2
   0         1         0         0
   0         0         0         1

T43
   0.5       0        -0.866     0
   0.866     0         0.5       0
   0        -1         0         0.82
   0         0         0         1

T54
   1         0         0         0
```

```
   0         0        -1         0
   0         1         0         0
   0         0         0         1

T65
   1         0         0         0
   0         1         0         0
   0         0         1         0.14
   0         0         0         1

T60
   0.8624   -0.07946  -0.5       0.06
  -0.3624   -0.7866   -0.5       0.06
  -0.3536    0.6124   -0.7071    0.277
   0         0         0         1

------------------------------------------------------------------------
Inverse Kinematics Solution Found: [ 0.48773184  0.66870682 -1.20807009
0.99950452  2.63022788 -1.67735574]
Inverse Kinematics Solution Found: [-2.84731863  2.03845298 -2.72631975
-1.74922772  1.42596723 -3.03108333]
Inverse Kinematics Solution Found: [-2.35626898 -3.08379134 -2.21498548
-3.14159178 -2.15718489  2.35626938]
```